

Enhancing intrusion detection performance using GCN-LOF: A hybrid graph-based anomaly detection approach

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ABSTRACT

To address limitations of traditional anomaly detection methods in handling high-dimensional sparse data and dynamic network environments, this study proposes an intrusion detection framework integrating Graph Convolutional Network (GCN) and Local Outlier Factor (LOF). Data is structured into graphs using k-nearest neighbors, and LOF scores quantify local density anomalies as additional node features, enhancing the model's sensitivity to rare and sparse attack patterns. GCN layers then extract global structural insights to improve detection performance. Empirical evaluations conducted on four benchmark intrusion detection datasets—Bot-IoT, CICIDS, UNSW-NB15, and ToN-IoT—demonstrate that the proposed model significantly improves detection accuracy, achieving a comprehensive F1 score of 0.964, representing an average increase of 5.3% compared to other state-of-the-art methods. Visualization techniques, including loss curves and t-SNE dimensionality reduction, substantiate the model's feature learning efficacy, while the detailed characterization of attack patterns and network protocols confirms its reliability and robustness in diverse intrusion detection scenarios.

1. Introduction

The rapid advancement of information and communication technologies has drastically increased the complexity and scale of modern network infrastructures, leading to an explosion of sophisticated cyber threats [1]. Intrusion Detection Systems (IDS) have become critical for ensuring network security by identifying attacks such as distributed denial-of-service (DDoS) [2], data exfiltration [3], and malicious code injection [4]. However, the ever-growing volume, high dimensionality, and dynamic nature of network traffic present significant challenges to traditional IDS approaches [5].

Existing IDS techniques—including rule-based systems, shallow machine learning [6], and deep learning—each exhibit notable limitations [7]. Signature-based and rule-driven methods require frequent updates and fail against zero-day threats [8], while conventional machine learning models such as support vector machines [9] and random forests [10] rely on handcrafted features and are often ineffective when dealing with sparse or high-dimensional data. Deep learning approaches, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs) [11,12], have enhanced feature representation capabilities, but remain computationally expensive and struggle to model structural relationships inherent in network traffic. Graph Convolutional Networks (GCNs) address this by capturing inter-entity dependencies within graph-structured data [13], yet they suffer

from over-smoothing effects and diminished performance in recognizing localized, sparse anomalies. Conversely, the Local Outlier Factor (LOF) algorithm [14] excels at identifying local density deviations but lacks the ability to model global structural context and is sensitive to high-dimensional noise.

These limitations underscore the necessity for a unified framework that can capture both global topological features and localized anomaly signals within dynamic and heterogeneous traffic environments. To this end, we propose a novel hybrid intrusion detection model GCN-LOF which, to the best of our knowledge, is the first to integrate the global feature learning of GCN with the local density estimation capability of LOF in a unified architecture. In our framework, network traffic data is first transformed into graph structures using the k-nearest neighbors (KNN) algorithm. LOF scores are then computed and embedded as auxiliary node attributes, introducing anomaly-aware local information directly into the feature space. These enriched node representations are subsequently processed through GCN layers, enabling the model to jointly leverage both global connectivity patterns and localized outlier characteristics.

The main contributions of this paper are summarized as follows:

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Table 1
Summary of abbreviations.

Abbreviation	Full form
GCN	Graph Convolutional Network
LOF	Local Outlier Factor
KNN	k-Nearest Neighbors
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
GNN	Graph Neural Network
F1	F1 Score
DoS	Denial of Service
DDoS	Distributed Denial of Service
IoT	Internet of Things

- We propose GCN-LOF, a novel hybrid intrusion detection framework that integrates graph-based learning and local density estimation to jointly capture global structural dependencies and local anomaly signals.
- We introduce a novel LOF-aware feature augmentation strategy, which embeds local outlier scores directly into node attributes. This enhances the anomaly sensitivity of GCN propagation, enabling effective detection of sparse and high-dimensional anomalies.
- Through extensive evaluation on four widely-used intrusion detection datasets—Bot-IoT, CICIDS, UNSW-NB15, and ToN-IoT—we show that GCN-LOF outperforms state-of-the-art methods by an average of 5.3% in F1-score, validating its generalization and superiority.

The remainder of this paper is organized as follows. Section 2 reviews related work in intrusion detection, focusing on graph-based and hybrid models. Section 3 introduces the proposed GCN-LOF framework. Section 4 presents the experimental setup and evaluation. Section 5 discusses limitations and future directions. Section 6 concludes the paper. A list of abbreviations is provided in Table 1.

2. Related work

Intrusion detection has evolved through various methodologies to address challenges like scalability, adaptability, and high-dimensional complexity in network security. Early systems relied on rule-based approaches, while recent advancements have shifted toward dynamic methods, including machine learning, deep learning, and graph-based techniques. This section reviews key works in five areas: rule-based systems, statistical models, machine learning, deep learning, and graph-based learning.

Early methods focused on rule-based intrusion detection systems (IDS), where domain knowledge was encoded into detection rules. Snort [15] identifies known threats using signature-based pattern matching. While interpretable, its reliance on static rule sets makes it ineffective against novel attacks. To improve adaptability, Wisanwanichthan and Thammawichai [16] introduced a Double-Layer Hybrid Approach (DLHA) that combines rule logic with statistical classification. However, hybrid rule-based systems struggle with scalability in large-scale traffic scenarios.

To address the rigidity of rule-based systems, researchers explored statistical anomaly detection. Breunig et al. [14] proposed the LOF, which detects anomalies by measuring local density deviation without labeled data. Abudurexiti et al. [17] employed unsupervised models to detect deviations in high-dimensional traffic, while Rodríguez et al. [18] developed a density estimation framework for evolving environments. However, these methods are sensitive to noise and rely on assumptions about feature distributions, which may not hold in real-world data.

With the rise of supervised learning, machine learning approaches improved detection accuracy. Rwbisira and Suchithra [19] employed Random Forest classifiers to analyze traffic datasets, showing better performance than rule-based systems. Similarly, SVM were used for binary classification tasks in intrusion detection [20]. While these models reduce reliance on expert rules, they often overlook structural dependencies or temporal context in cyberattacks.

Deep learning techniques further advanced the field by enabling automatic feature extraction. Kim et al. [21] applied CNN to detect spatial anomalies in traffic data. LSTM and GRU-based models, such as those by Cho and Cuong [22] and Zhou et al. [23], have proven effective for modeling long-term dependencies. However, these models treat data points independently and do not capture relational structures among network entities, limiting their effectiveness for distributed or coordinated attacks.

Graph-based learning methods have emerged to address this structural limitation. Sun et al. [24] proposed GNN-IDS, which leverages GNN to learn from node features and traffic topology. Duan et al. [25] introduced a dynamic spatiotemporal GCN that models temporal evolution and inter-node dependencies. More recent efforts focus on enhancing representation capacity and anomaly awareness. Li et al. [26] proposed SEGA-Net, a semantic-enhanced graph attention model. Latif-Martínez et al. [27] proposed GAT-AD, a graph-based anomaly detection model that leverages attention mechanisms to learn contextual dependencies among OD flows and identify temporal anomalies in network traffic. Goode et al. [28] proposed LUNAR, a hybrid model that integrates local outlier detection with GNNs to improve anomaly detection on tabular data. Despite these advances, most graph-based approaches treat density estimation as a peripheral component.

Despite progress in graph-based and hybrid methods, a unified framework that simultaneously models local anomaly patterns and global structural relationships in network traffic remains lacking. Existing models either separate anomaly detection from graph propagation or treat outlier scoring as a post-processing step. This paper proposes GCN-LOF, a hybrid framework that integrates LOF-based local density estimation into graph convolutional learning. By embedding anomaly-aware scores into node features and incorporating them during neighborhood aggregation, the model enables joint reasoning over localized deviations and graph topology, enhancing detection of rare, distributed, and high-dimensional anomalies in dynamic intrusion scenarios.

3. Method

3.1. Problem definition

Traditional methodologies often struggle to address the challenges of underreporting anomalous instances in high-dimensional feature spaces, leading to low detection rates for rare or evolving attack patterns. Additionally, accurately modeling relationships within network traffic in dynamic environments remains a significant challenge, as many existing approaches fail to leverage both local and global structural dependencies effectively. These limitations hinder the detection accuracy, robustness, and generalization of intrusion detection systems.

To overcome these challenges, this paper introduces an intrusion detection model designed to enhance detection performance by integrating GCN with the LOF method. The proposed approach improves anomaly sensitivity and robustness by leveraging graph-structured data to capture network-wide correlations while incorporating local density variations to highlight rare attacks.

The network traffic data is represented as feature vectors $\mathbf{X} \in \mathbb{R}^{N \times d}$, where N is the number of samples, d is the feature dimension, and the sample labels $y \in \{0, 1\}$ indicate normal and anomalous traffic. The adjacency matrix $\mathbf{A}$ is constructed using the KNN algorithm, which maps traffic samples into a graph structure, preserving connectivity information. The LOF algorithm quantifies the local density of each

sample, aiding in the identification of sparse anomalies that traditional models may overlook.

Importantly, the integration of GCN and LOF leverages their complementary strengths. While GCN excels at capturing global structural patterns through neighborhood aggregation, it tends to oversmooth node representations, especially in scenarios with limited anomalous instances. LOF, on the other hand, is specifically designed to detect local density deviations, making it highly sensitive to isolated or sparsely distributed anomalies. By embedding LOF scores into node features and influencing graph propagation through anomaly-aware weighting, the proposed model amplifies the impact of locally irregular samples within the global graph context. This hybrid mechanism enables the detection system to simultaneously preserve structural semantics and emphasize subtle deviations, thereby improving its capacity to detect both clustered and isolated intrusion behaviors.

The model constructs an enhanced feature matrix $\mathbf{F}_{\text{enhanced}}$, optimizes node embeddings $\mathbf{H}$ using GCN layers, and classifies network traffic through a trained anomaly detection module. The ultimate objective is to minimize the classification loss function $\mathcal{L}_{\text{classification}}$, thereby improving detection accuracy, reducing false positives, and increasing resilience against dynamic network anomalies.

3.2. Model architecture

This paper presents an anomaly detection framework based on GCN and LOF, as illustrated in Fig. 1. The model takes raw data instances as input and outputs anomaly predictions through a three-stage process that integrates local outlier modeling with graph-based learning. Specifically, each data sample is first encoded into a node feature vector, and an LOF score is computed to characterize its local density deviation. The LOF score is then appended to the original feature, forming an enhanced feature vector whose construction is detailed in the following subsection. This enriched representation incorporates anomaly-sensitive information that supports downstream graph modeling.

Based on the enhanced node features, a graph structure is constructed using the KNN algorithm, where edges represent neighborhood relations in the feature space. This step transforms the input data into a graph format that captures both semantic similarity and structural locality.

Finally, a graph convolutional network is applied to the constructed graph, where message passing is guided by both feature similarity and LOF-based weighting. This design introduces explicit anomaly awareness into the propagation process, enabling the model to focus on structurally significant and statistically deviant nodes. By combining local density estimation with global relational modeling, the proposed GCN-LOF framework enhances the detection of subtle and irregular anomalies in complex data environments.

3.2.1. Feature vector construction from intrusion data

Before constructing the anomaly-aware graph, raw intrusion detection data undergo a preprocessing pipeline that transforms heterogeneous network traffic records into structured node feature vectors. For each dataset, we first exclude non-informative or identifier fields such as IP addresses, port numbers, and attack types, and retain only numerical attributes that contribute to anomaly characterization. These retained features are normalized using Min-Max scaling to ensure consistency in scale across all dimensions.

To enhance anomaly awareness, we compute the LOF for each sample using a k -nearest neighbor approach. The LOF score reflects the local density deviation of a point relative to its neighborhood, enabling the model to capture fine-grained statistical irregularities. These LOF scores are appended as an additional dimension to the normalized feature vectors, resulting in an enriched node representation matrix $X' \in \mathbb{R}^{N \times (d+1)}$. This matrix integrates both semantic feature information and localized anomaly signals, serving as the foundation for subsequent graph construction.

3.2.2. Anomaly-aware graph construction

To construct a graph structure that is sensitive to local anomalies, we leverage the LOF-augmented feature matrix X' and perform graph construction using the KNN algorithm. Let the input dataset be represented by $X \in \mathbb{R}^{N \times d}$, where N is the number of samples and d is the number of original features. After appending the LOF score, each node feature vector becomes $(d+1)$ -dimensional.

We apply the k -nearest neighbors algorithm to X' using Euclidean distance as the similarity metric to identify local neighborhood relations. The resulting adjacency matrix $A \in \{0, 1\}^{N \times N}$ is defined such that an edge $A[i, j] = 1$ exists if node j belongs to the set of k nearest neighbors of node i in the $(d+1)$ -dimensional feature space. To interface with the graph neural network, we convert the adjacency matrix into coordinate list format $\text{edge_index} \in \mathbb{N}^{2 \times E}$, compatible with PyTorch Geometric (PyG). The resulting graph object $\mathcal{G} = (\text{edge_index}, X')$ serves as input to the graph convolutional model.

3.2.3. LOF-Guided graph convolutional propagation

To exploit both the structural topology and anomaly relevance of the graph, we introduce a LOF-guided graph convolutional network. Let $H^{(0)} = X'$ denote the initial node features. At each layer l , node embeddings are updated via a weighted message-passing scheme that considers both feature similarity and anomaly intensity.

The propagation weight from node j to node i is defined as:

$$w_{ij} = \exp \left(-\alpha \|H_i^{(l)} - H_j^{(l)}\|^2 \right) \cdot (1 + \beta \cdot \text{LOF}(z_j)), \quad (1)$$

where α and β are hyperparameters controlling the influence of similarity and anomaly emphasis, respectively. This weighting strategy assigns greater importance to neighbors that are both semantically close and statistically deviant.

The updated embedding for node i is computed as:

$$H_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} w_{ij} \cdot H_j^{(l)} W^{(l)} \right), \quad (2)$$

where $W^{(l)}$ is a learnable weight matrix and $\sigma(\cdot)$ denotes a ReLU activation function. This design enables dynamic adjustment of information flow based on local irregularity.

The GCN-LOF model adopts a two-layer architecture. The first layer projects the augmented features to a hidden dimension h , and the second layer maps h to a scalar logit, followed by a sigmoid activation for anomaly probability prediction.

The model is optimized using binary cross-entropy loss:

$$\mathcal{L} = \sum_{i=1}^N \left[-y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i) \right] + \lambda \|W^{(1)}\|^2, \quad (3)$$

where $y_i \in \{0, 1\}$ is the ground-truth label, $\hat{y}_i$ is the predicted probability, and λ is the ℓ_2 regularization coefficient. This objective ensures that both semantic structure and statistical deviations are effectively captured for robust anomaly detection.

4. Experiments

4.1. Dataset

To comprehensively evaluate the anomaly detection capability of the proposed GCN-LOF model, we conduct experiments on four widely adopted intrusion detection datasets: Bot-IoT [29], CICIDS [30], UNSW-NB15 [31], and ToN-IoT [32]. These datasets reflect diverse network environments and incorporate both benign and malicious traffic, encompassing a variety of attack types such as DDoS, DoS, probing, malware, and information theft.

Bot-IoT is a NetFlow-style dataset generated by extracting flow-level features from PCAP files, with each flow labeled by attack category. It includes four attack types and exhibits an extremely imbalanced distribution, with over 99% of flows classified as anomalous. CICIDS2017

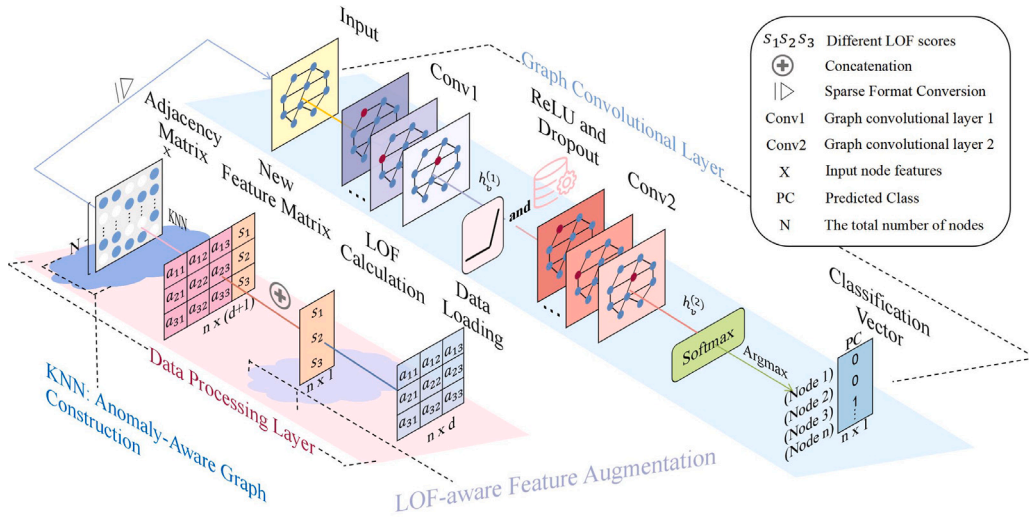


Fig. 1. Model architecture of GCN-LOF. The framework includes LOF-based anomaly enhancement, KNN-based graph construction, and LOF-weighted graph convolutional propagation.

Table 2
Dataset statistical information.

Dataset	Total	Features	Normal	Anomalous
Bot-IoT	1,000,000	45	3569	996,431
CICIDS	1,000,000	45	880,623	119,377
UNSW-NB15	1,000,000	45	960,078	39,922
ToN-IoT	1,000,000	45	359,618	640,382

captures realistic multi-day traffic under various protocols and attack scenarios, with 78 features extracted per flow using CICFlowMeter. UNSW-NB15 contains synthetic traffic collected in a cyber range, comprising normal and nine attack types, each sample represented by 45 network features. ToN-IoT integrates telemetry from IoT sensors, network flows, and host logs, and covers a wide spectrum of attacks across multiple data modalities.

Each dataset exhibits distinct characteristics. Bot-IoT and ToN-IoT present heavy class imbalance with dense attack patterns, while CICIDS and UNSW-NB15 are relatively more balanced but include controlled or emulated attacks. These differences are critical when interpreting cross-dataset performance variations, as summarized in Table 2.

To further illustrate the composition of the datasets, Fig. 2 visualizes the distribution of major traffic categories across each dataset. Here, “Normal” refers to benign traffic, while the remaining bars represent three major attack types: DoS, DDoS, and Reconnaissance. Bot-IoT and ToN-IoT exhibit attack-heavy distributions dominated by DDoS and reconnaissance traffic, respectively. CICIDS contains primarily normal traffic with a moderate amount of DoS and DDoS, while UNSW-NB15 presents mostly benign traffic and includes no DDoS samples. These differences highlight the diversity of anomaly structures, providing varied evaluation scenarios for the proposed model.

Considering the aforementioned differences in traffic composition, feature distributions, and attack semantics, we further construct a merged dataset to evaluate the model’s generalization capability under heterogeneous and cross-domain conditions. To mitigate domain shift and ensure feature consistency, we retain only the 45 numerical features shared across all datasets and remove identifier-related fields such as IP addresses and port numbers. Z-score normalization is applied independently to each dataset to preserve intra-domain statistical properties while maintaining cross-dataset comparability. Additionally, LOF scores are computed separately for each dataset (with $k = 10$) and appended as an auxiliary feature, capturing local density deviations specific to each domain. After stratified splitting into 80% training and 20% testing subsets, the datasets are concatenated to form the final

unified training and testing sets. This preprocessing pipeline allows the model to jointly learn from a wide variety of network behaviors and anomaly patterns, while effectively addressing inter-dataset discrepancies without introducing artificial balancing or domain adaptation components. The model’s consistent performance on this unified dataset confirms the practical value of the merging strategy.

4.2. Experimental setup

To ensure reproducibility and fairness in performance evaluation, all features are standardized using Z-score normalization. The datasets do not contain missing values and therefore require no additional imputation or cleaning. Each dataset is randomly partitioned into 80% for training and 20% for testing, with stratified sampling applied to preserve the original class distribution. A fixed random seed is used throughout the experiments to ensure consistent data splits and results across runs.

Given the pronounced class imbalance in datasets such as Bot-IoT, where the ratio of normal to abnormal samples is extremely skewed, class weights are adjusted inversely proportional to class frequencies. This weighting strategy emphasizes learning from underrepresented instances and reduces bias toward the dominant class, thereby improving the model’s sensitivity to rare but critical anomalies.

The model is trained using the Adam optimizer with a learning rate of 0.01 and weight decay of 5×10^{-4} . The embedding dimension is set to 16, and training is conducted over 50 epochs. A two-layer GCN architecture is adopted, which balances expressive capacity and training stability. This design follows prior evidence that deeper GCNs often suffer from over-smoothing, limiting their ability to preserve localized feature distinctions critical for anomaly detection [33].

The k value used for constructing the k -nearest neighbor (KNN) graph and for calculating LOF scores is a key hyperparameter. Based on prior literature and the expected data sparsity in anomaly detection tasks, we hypothesize that moderate k values strike a better trade-off between preserving local density structures and avoiding noise amplification. We initially set $k = 10$ and empirically evaluated its sensitivity by varying $k \in \{5, 10, 20, 30\}$, as summarized in Table 3. Smaller k values such as $k = 5$ favor precision by detecting concentrated anomalies but suffer from low recall. Larger values like $k = 20$ or $k = 30$ increase recall but reduce precision due to neighborhood dilution. The optimal trade-off is achieved at $k = 10$, which consistently delivers the highest F1 across datasets.

To further evaluate model robustness, we test sensitivity to different random seeds. As shown in Table 4, variations in seed values result in

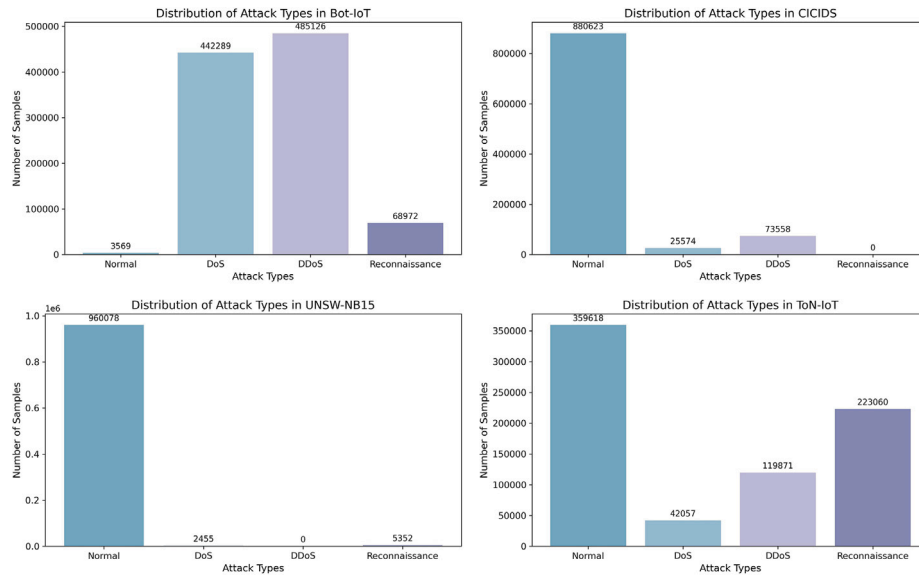


Fig. 2. Distribution of attack types across four datasets: Bot-IoT, CICIDS, UNSW-NB15, and ToN-IoT. “Normal” denotes benign traffic; the remaining categories represent DoS, DDoS, and Reconnaissance attacks.

Table 3
Impact of different k values on GCN-LOF performance.

k	ACC	PRE	REC	F1
5	0.919	0.904	0.780	0.834
10	0.957	0.906	0.866	0.884
20	0.916	0.851	0.881	0.859
30	0.939	0.880	0.863	0.865

Table 4
Effect of different random seeds on GCN-LOF performance.

Random seed	ACC	PRE	REC	F1
1	0.955	0.901	0.834	0.860
10	0.946	0.936	0.682	0.696
20	0.926	0.893	0.823	0.856
30	0.953	0.900	0.840	0.865
40	0.949	0.902	0.845	0.870
42	0.957	0.906	0.866	0.884

only marginal fluctuations in performance metrics. This demonstrates that the model’s learning process is stable and not overly sensitive to initialization, underscoring the consistency and reliability of the proposed training strategy.

Additionally, the LOF-specific parameters `min_samples` and `radius` are set to 5 and 0.5, respectively. These values are consistent with established practices for density-based anomaly detection in high-dimensional settings [34], providing a balance between local sensitivity and global robustness. Notably, these LOF parameters remain fixed across all datasets and experiments without any dataset-specific tuning, yet consistently yield strong performance. This design choice highlights the model’s robustness and practicality in real-world deployment scenarios (see Table 5).

To ensure cross-model consistency, we additionally maintain a unified configuration baseline for all compared methods, including Isolation Forest, DeepAE, EFC, and LOF. While only a small set of parameters—namely `DATA_PATH`, `SEED`, and `silos`—are universally applicable across all models, this reflects the architectural heterogeneity and implementation-specific requirements inherent to these baselines. For reproducibility, model-specific parameters are retained using default values or settings recommended in previous literature. This strategy balances compatibility with fairness and allows each

Table 5
Experimental and hyperparameter settings for GCN-LOF.

Item	Setting
Libraries	PyTorch 1.12.0, CUDA 11.7
Normalization	Z-score
Distance metric	Euclidean
LOF parameters	Min Samples = 5, Radius = 0.5
Batch size	128
Hidden dimension	16
Learning rate	0.01
Weight decay	5e−4
Epochs	50
k (Neighbors)	10
Optimizer	Adam
Loss function	NLL

Table 6
Summary of hyperparameter configurations used in baseline methods.

Parameter	Default	Used by	Purpose
<code>DATA_PATH</code>	<code>datasets/</code>	All	Dataset directory path
<code>SEED</code>	42	All	Randomness control
<code>silos</code>	Dynamic	All	Data split structure
<code>BATCH_SIZE</code>	128	DeepAE	Mini-batch size
<code>EPOCHS</code>	10	DeepAE	Training epochs
<code>VAL_RATIO</code>	0.1	DeepAE	Validation split ratio
<code>TH_QTL</code>	0.95	DeepAE, EFC	Threshold for error/energy
<code>contamination</code>	Not set	IsoForest	Outlier ratio (auto-estimated)
<code>n_neighbors</code>	20	LOF	LOF neighborhood size
<code>novelty</code>	True	LOF	Enable test-time prediction
<code>cutoff_quantile</code>	0.95	EFC	Energy score threshold

model to operate under its standard assumptions while sharing a common data processing pipeline. A summary of shared and model-specific parameters is shown in Table 6.

In summary, the experimental setup confirms both the empirical robustness and adaptability of the proposed GCN-LOF model. The inclusion of standard training procedures, sensitivity analyses, and baseline alignment strategies provides a solid foundation for subsequent comparative evaluation and ablation studies.

4.3. Comparative evaluation

To validate the effectiveness and generalizability of the proposed GCN-LOF model, we compare it against six representative baseline

Table 7
F1 scores of different models on four datasets.

Model	Bot-IoT	CICIDS	UNSW-NB15	ToN-IoT
IsoForest	0.007	0.924	0.949	0.579
DeepAE	0.224	0.750	0.511	0.210
EFC	0.974	0.660	0.620	0.660
LOF	0.014	0.935	0.973	0.723
E-ResGAT	0.873	0.817	0.825	0.677
GNN-NIDS	0.539	0.880	0.833	0.649
GCN-LOF	0.946	0.945	0.982	0.982

methods: Isolation Forest (IsoForest) [35], Deep Autoencoder (DeepAE) [36], Efficient Feature Clustering (EFC) [37], Local Outlier Factor (LOF) [38], E-ResGAT [39], and GNN-NIDS [40]. These methods span tree-based, neural network-based, clustering-based, density-based, and graph-based paradigms, offering a comprehensive benchmark framework.

Each baseline possesses distinct strengths and limitations. IsoForest excels in linear-time scalability but performs poorly in high-dimensional or sparse scenarios due to its reliance on recursive partitioning. DeepAE captures global structures in feature space via reconstruction loss, but its sensitivity to subtle or sparsely distributed anomalies is limited. EFC leverages clustering for anomaly localization but assumes feature consistency, which may not hold in dynamic traffic. Standard LOF, while effective in modeling local irregularities, suffers from scalability constraints and lacks integration with global structural representations. E-ResGAT and GNN-NIDS incorporate graph attention or relational inductive bias, respectively, but rely primarily on static node attributes and often lack explicit density-awareness, limiting their adaptability to mixed-distribution scenarios.

Table 7 summarizes the F1-scores achieved by each model across four benchmark datasets. GCN-LOF consistently outperforms all baselines across the board. Notably, on Bot-IoT, where anomaly instances are densely concentrated and normal traffic is minimal, GCN-LOF achieves an F1-score of 0.946. This substantially exceeds DeepAE (0.224) and IsoForest (0.007), reflecting GCN-LOF's robustness in identifying dense, high-magnitude anomalies while avoiding trivial overfitting to the dominant class. The LOF baseline struggles with scalability, capturing only limited patterns ($F1 = 0.014$), which further highlights the advantage of fusing local density with graph-level propagation. E-ResGAT and GNN-NIDS exhibit moderate performance on Bot-IoT ($F1 = 0.873$ and 0.539), suggesting some capacity for structural modeling but limited anomaly discrimination under extreme imbalance.

In contrast, on CICIDS, where traffic exhibits clearer patterns and better class balance, GCN-LOF still demonstrates superior performance ($F1 = 0.945$), slightly outperforming LOF (0.935). This result suggests that even in structured environments, the hybrid architecture of GCN-LOF enables it to maintain sensitivity to both localized and global variations in feature distributions. GNN-NIDS performs competitively ($F1 = 0.880$), while E-ResGAT reaches 0.817, indicating that graph-based models benefit from structural patterns but may lack robustness in differentiating fine-grained attack semantics.

GCN-LOF also maintains consistent superiority on UNSW-NB15 and ToN-IoT, two datasets characterized by complex background traffic and mixed attack types. Its high F1-scores (0.982 for both) reflect the model's capacity to reconcile feature space heterogeneity with structural pattern learning, suggesting that the fusion of LOF-enhanced node features with GCN propagation yields strong generalizability. In comparison, both E-ResGAT and GNN-NIDS achieve stable but lower scores (around 0.82–0.83), underscoring the value of incorporating anomaly-aware features into the graph framework.

To assess generalization under diverse conditions, we further constructed a comprehensive dataset by merging the four individual datasets. This mixed dataset simulates real-world traffic complexities, including heterogeneous protocols, imbalanced class distributions, and

Table 8
Performance of models on the comprehensive dataset.

Model	Recall	Accuracy	F1 Score
IsoForest	0.894	0.608	0.615
DeepAE	0.300	0.607	0.325
EFC	0.851	0.879	0.768
LOF	0.931	0.784	0.661
E-ResGAT	0.758	0.900	0.798
GNN-NIDS	0.772	0.887	0.725
GCN-LOF	0.931	0.965	0.964

Table 9
GCN-LOF performance per class on four datasets.

Dataset	Class	Precision	Recall	F1-score
Bot-IoT	0	0.833	0.821	0.841
	1	0.997	1.000	0.998
CICIDS	0	0.971	0.987	0.979
	1	0.889	0.783	0.833
UNSW-NB15	0	0.984	0.997	0.990
	1	0.885	0.615	0.726
ToN-IoT	0	0.705	0.748	0.726
	1	0.854	0.824	0.839

varying anomaly densities. Table 8 reports model performance under this unified setting.

Traditional models, including IsoForest and DeepAE, exhibit sharp performance degradation in this context ($F1 = 0.615$ and 0.325 , respectively), as their mechanisms are unable to capture cross-domain feature inconsistencies. LOF improves recall but sacrifices precision, indicating over-sensitivity to minor deviations. EFC, E-ResGAT, and GNN-NIDS offer improved balance, with F1-scores ranging from 0.725 to 0.798, yet still fall short of handling multi-source anomaly variance. In contrast, GCN-LOF achieves a high F1-score of 0.964, combining 0.931 recall with 0.965 accuracy. This performance stems from its dual ability to model long-range structural dependencies via GCN and capture fine-grained local irregularities through LOF, allowing it to distinguish true anomalies from benign variance across domains.

To further examine GCN-LOF's resilience to class imbalance, we evaluate per-class metrics across the four datasets. As shown in Table 9, GCN-LOF achieves near-perfect detection of attack traffic in Bot-IoT ($F1 = 0.998$, Recall = 1.000), while maintaining a reasonable performance for normal traffic ($F1 = 0.841$), despite the overwhelming imbalance (attack:normal $\approx 300:1$). In datasets such as UNSW-NB15 and ToN-IoT, the model still preserves discrimination capability across both classes, reflecting its balanced representation learning. This suggests that the integration of anomaly scores into the graph structure enables GCN-LOF to capture minority-class patterns without oversampling or heuristic balancing.

In summary, GCN-LOF not only outperforms established baselines across diverse datasets, but also demonstrates strong adaptability to heterogeneity in both traffic distribution and anomaly manifestation. Its hybrid architecture allows it to integrate multi-level information, enabling fine-grained discrimination in highly imbalanced and structurally complex environments.

4.4. Attack and protocol analysis

To assess the GCN-LOF model's capability in capturing diverse anomaly patterns, we investigate its response to different attack types and protocol behaviors. This analysis evaluates how well the model adapts to various network attack strategies and how protocol-specific traffic patterns influence the detection of anomalies.

Fig. 3 visualizes the network traffic knowledge graph for different attack types, showing source–destination relationships, port usage, and connection patterns associated with each attack. For Denial-of-Service (DoS) attacks, the model detects highly centralized graph structures

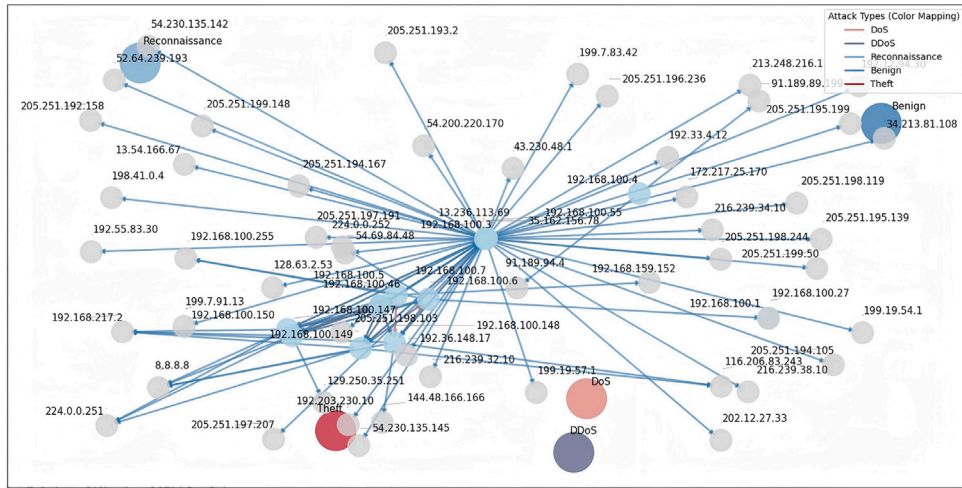


Fig. 3. Network traffic knowledge graph visualizing different attack types. The diagram captures source-destination relationships, port usage, and attack-specific connection patterns.

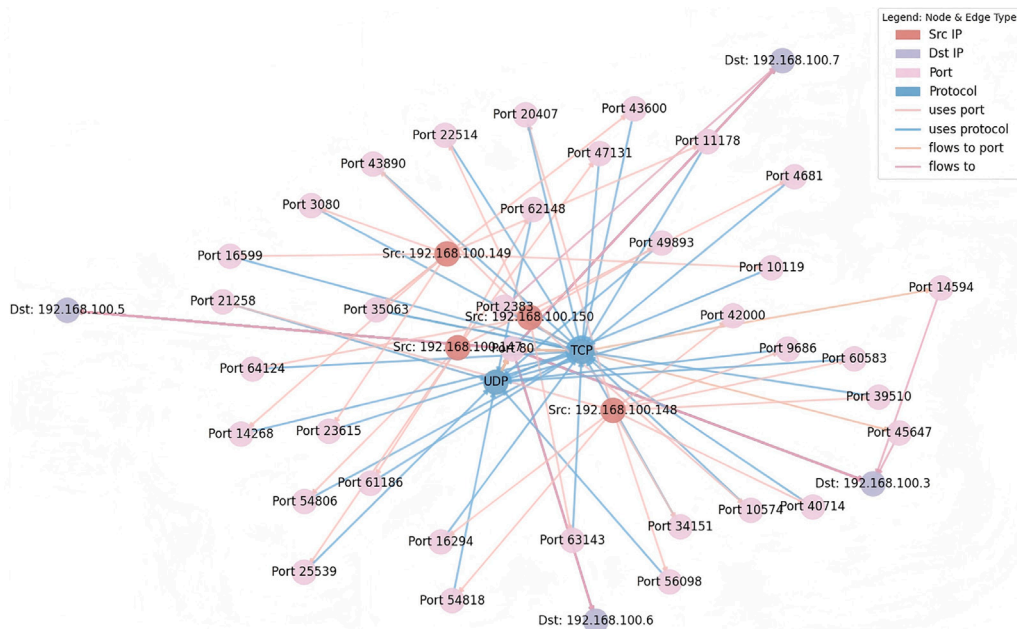


Fig. 4. Visualization of network connections categorized by protocol type. The figure highlights protocol-specific anomalies and their role in different attack strategies.

due to a single source flooding a target with frequent connections. Distributed DoS (DDoS) attacks, originating from multiple sources, exhibit more dispersed and less centralized connection patterns. In contrast, reconnaissance attacks generate sparse, low-intensity scans, leading to globally distributed anomalies.

These attack patterns are further analyzed through protocol-level behaviors. The model is able to distinguish anomalies in TCP, UDP, and ICMP protocols based on their distinctive connection characteristics. TCP-based anomalies, such as SYN Floods, often result in incomplete handshakes due to the connection-oriented nature of the protocol. UDP-based attacks, by contrast, lack session states and generate bursty, unidirectional flows, which are harder to trace but can still be identified by analyzing density deviations. ICMP-based anomalies, while structurally simpler, are identified by irregular packet sizes or isolated flow patterns.

Fig. 4 shows the visualization of network connections categorized by protocol type. For TCP-based anomalies, the model identifies concentrated flows that occur during incomplete handshakes. These connections are concentrated in the initial phases of communication, where

the model detects the absence of expected protocol behavior, such as a missing second handshake. In the case of UDP, the model detects burst-like patterns that lack connection state information, leading to irregular, one-way flows across the network. These bursts are often sporadic and scattered, with the model recognizing these flows as anomalous due to density deviation compared to regular traffic. For ICMP-based anomalies, the visualization highlights isolated nodes or sparse edges that occur due to irregular packet sizes or unusual communication patterns. These anomalies are typically more difficult to trace since ICMP lacks the inherent structure of TCP/UDP, but they are still detectable by the model through their statistical deviation from normal packet distributions. In particular, the graph-based encoding enables the model to capture protocol-specific communication dynamics, such as local density shifts and interaction sparsity, which are often overlooked by purely statistical methods.

This detailed analysis of protocol-specific anomalies illustrates the GCN-LOF model's capacity to differentiate between attack behaviors based on the underlying transport mechanisms. By encoding these protocol-specific variations into the graph representation, the model

Table 10

Ablation study results: The impact of different components on GCN-LOF performance (averaged across four datasets).

Dropout	Hidden layer	LOF	GCN	k	Accuracy	Precision	Recall	F1 score
✓	✓	✓	✓	10	0.957	0.906	0.866	0.884
✓	✓	✓	✓	10	0.889	0.632	0.691	0.650
✓	✓	✓	✓	10	0.870	0.595	0.500	0.445
✓	✓	✓	✓	20	0.916	0.851	0.881	0.858
✓	✓	✓	✓	10	0.968	0.719	0.659	0.681
✓	✓	✓	✓	10	0.898	0.758	0.902	0.771

enhances its anomaly detection performance, making it suitable for real-world network defense applications.

4.5. Ablation study

To investigate the contribution of each major component in the GCN-LOF architecture, we conduct a series of ablation experiments by systematically removing or modifying specific elements while keeping others fixed. The evaluated variants include the exclusion of dropout regularization, the removal of the hidden layer in the GCN, the elimination of LOF-based features, the replacement of GCN with a GraphSAGE module, and the variation of the neighborhood size k in the LOF computation. The average results across four benchmark datasets are summarized in Table 10.

The first observation concerns the effect of dropout regularization. When dropout is disabled, the F1-score drops markedly from 0.884 to 0.650, indicating that the model becomes more prone to overfitting, especially in scenarios with sparse anomalies. This decline is accompanied by reduced precision and recall, suggesting a degradation in both discriminative ability and anomaly sensitivity. Dropout serves as an essential regularization mechanism that introduces randomness during training, encouraging the network to learn more generalizable patterns rather than memorizing majority-class features.

The hidden layer within the GCN plays a particularly critical role. Its removal leads to the most substantial performance degradation, with the F1-score plummeting to 0.445. This sharp decline highlights the insufficiency of shallow transformations in capturing structural dependencies within graph-encoded traffic. The hidden layer facilitates multi-hop information propagation, enabling the model to integrate broader contextual information and produce more expressive node embeddings. Without it, the GCN is limited to linear mappings, which weakens its ability to represent complex interactions between nodes.

The contribution of LOF-based features is also evident. When these features are excluded, the model exhibits a clear decline in both precision and recall, with the F1-score reduced to 0.681. This indicates that the LOF module enhances the model's sensitivity to subtle or localized anomalies by injecting local density information into the learning process. Without this augmentation, the model lacks fine-grained spatial awareness and is less effective in distinguishing anomalous patterns from noisy but normal traffic behavior.

Replacing the GCN with a GraphSAGE module results in a modest drop in F1-score (from 0.884 to 0.771) and accuracy, despite a slight increase in recall. This suggests that while GraphSAGE retains some ability to model neighborhood structure, its aggregators are less expressive than those in GCN for capturing fine-grained graph dependencies. Consequently, the fusion of GraphSAGE with LOF-based features does not achieve the same level of synergy as the original design, leading to reduced performance on complex traffic patterns.

Adjusting the LOF neighborhood size k from 10 to 20 leads to a moderate increase in recall (from 0.866 to 0.881), indicating improved sensitivity to dispersed anomalies. However, this gain comes at the cost of decreased precision (from 0.906 to 0.851), reflecting a higher false positive rate. This trade-off is consistent with the behavior of LOF: larger k values smooth the local density landscape, potentially

obscuring sharp deviations associated with isolated anomalies, whereas smaller values yield more focused but potentially noisier estimations.

Complementing these metric-based findings, the training dynamics reveal further insights. The full GCN-LOF model consistently demonstrates rapid and stable convergence across datasets, whereas configurations lacking the hidden layer or LOF features exhibit slower optimization and higher final loss. In particular, the no-dropout variant tends to converge prematurely to a suboptimal plateau, underscoring its limited generalization. These trends are visualized in Fig. 5, which illustrates the test loss curves under different module configurations.

In summary, the ablation study demonstrates that each component of the GCN-LOF architecture is essential to its effectiveness. Dropout mitigates overfitting and enhances generalization, the hidden layer enriches structural expressiveness, and the LOF module introduces local density awareness critical for anomaly sensitivity. The comparison with GraphSAGE confirms the importance of strong graph aggregation mechanisms in preserving fine-grained structural cues vital to robust anomaly detection.

4.6. Sensitivity analysis of propagation hyperparameters

The propagation stage in GCN-LOF introduces two adjustable coefficients, α and β , which influence the relative weight of feature similarity and local density information during neighborhood aggregation. Specifically, α controls the strength of similarity-based propagation, while β adjusts the influence of nodes with high LOF scores. These two factors jointly shape how the model balances structural coherence with anomaly sensitivity in graph-based detection.

We test combinations of $\alpha \in \{0.1, 0.5, 1.0\}$ and $\beta \in \{0.0, 0.5, 1.0\}$ on four benchmark datasets. The results are illustrated in Fig. 6 using grouped bar charts, where each bar represents the F1-score corresponding to a specific (α, β) pairing. Compared with fixed baselines, the model exhibits clear sensitivity to both parameters. When $\beta = 0$, the propagation process ignores LOF weighting and degrades to standard similarity-based aggregation, resulting in relatively moderate but stable F1-scores. As β increases, the model becomes more responsive to local anomaly signals, which improves recall and enhances detection performance, particularly in datasets with sparse or subtle attack patterns.

The effect of α is also notable. A low value weakens neighborhood similarity, causing unstable propagation, while a high value overly restricts information flow from dissimilar neighbors. In contrast, $\alpha = 0.5$ consistently leads to more balanced results. Overall, the best performance is achieved when $\alpha = 0.5$ and $\beta = 1.0$, suggesting that moderate structural guidance combined with anomaly-aware weighting yields optimal results.

These findings confirm the utility of the LOF-weighted propagation mechanism and demonstrate that proper hyperparameter selection can effectively enhance the model's adaptability across diverse network conditions.

5. Discussion

5.1. Performance and limitations

The experimental results demonstrate that GCN-LOF effectively integrates graph structural learning and local anomaly detection, yielding high detection performance across diverse network environments. Particularly in imbalanced datasets such as Bot-IoT, the model achieves superior recall and precision due to the sensitivity introduced by LOF-based density signals. This highlights its robustness in identifying sparse and low-frequency intrusions—a notable challenge in intelligent intrusion detection.

However, in more heterogeneous datasets like ToN-IoT, performance is adversely affected by overlapping feature distributions and dispersed anomaly patterns. This degradation is reflected in higher

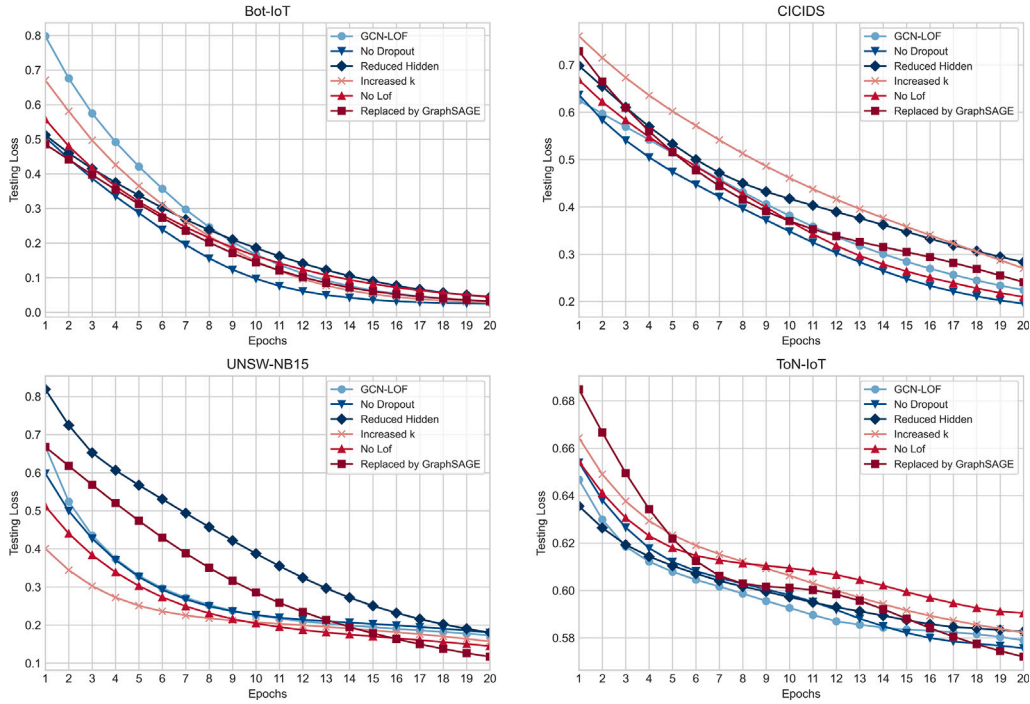


Fig. 5. Ablation study results of the GCN-LOF model, illustrating the impact of different module configurations on detection performance. The reported metrics are averaged across four datasets.

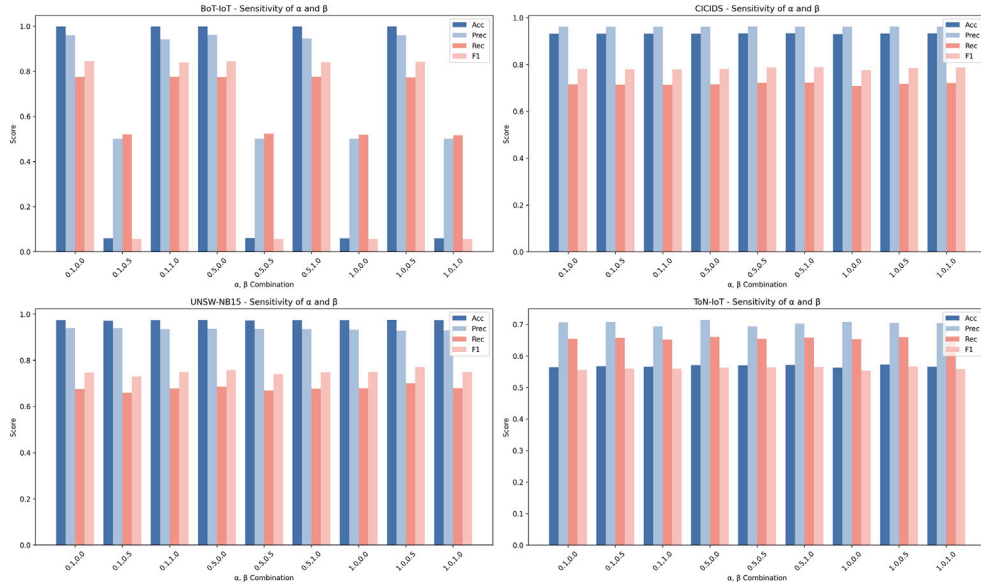


Fig. 6. Performance of GCN-LOF across four datasets under varying combinations of propagation hyperparameters α and β . The bar charts display four evaluation metrics—Accuracy, Precision, Recall, and F1-score—highlighting the impact of different parameter settings on detection effectiveness.

validation losses and less distinct class separation in t-SNE embeddings. We attribute this to diluted local density variations, which compromise the effectiveness of LOF. For example, reconnaissance attacks and UDP-based anomalies tend to resemble benign behaviors in certain statistical dimensions, making them harder to separate using local or global modeling alone. Addressing this issue may require incorporating temporal flow aggregation, adaptive neighborhood graph construction, or meta-learning techniques to improve the model's ability to generalize across variable attack types and traffic distributions [41].

5.2. Interpretability and future work

Although GCN-LOF achieves strong empirical results, its interpretability remains limited—an important concern for high-stakes deployment scenarios such as network forensics or automated response systems. Current visualizations, such as t-SNE embeddings and loss curves, offer coarse-level insights but cannot explain specific model decisions.

Table 11
System resource usage during GCN-LOF inference.

Metric	Value	Note
Total processes	12	Active processes
CPU usage (user)	21.1%	Time in user space
CPU idle time	78.6%	Idle CPU time
Memory used	24,385.3 MiB (~24 GiB)	RAM used by model
Free memory	441,418.9 MiB (~431 GiB)	Free RAM
Buffer/Cache	50,012.1 MiB (~49 GiB)	Cache/buffer memory
Swap usage	3547.3 MiB (~3.5 GiB)	Swap used
Available memory	488,217.3 MiB (~476 GiB)	Total usable memory

Notably, the LOF-based dynamic weighting mechanism in GCN-LOF introduces a built-in interpretability prior: nodes with higher local deviation (LOF scores) exert greater influence during message passing. This anomaly-guided propagation helps highlight locally irregular regions in the graph, suggesting that the model's focus aligns with statistical outliers. However, this mechanism alone is insufficient for fine-grained or instance-level explanation.

To address this, future work will integrate dedicated interpretability frameworks. GNNExplainer [42] can be used to extract node-level subgraphs and feature attributions responsible for anomaly decisions, while SHAP [43] may help quantify feature-wise contributions—including LOF—to the model's output. Additionally, incorporating an attention mechanism into the GCN layers (e.g., via GAT [44]) could enable visual tracking of important neighbors during propagation. These methods not only improve model transparency but also facilitate post-hoc auditability and trust, making GCN-LOF more suitable for operational environments requiring human-in-the-loop verification.

5.3. Scalability and deployment challenges

Despite promising results, GCN-LOF faces several scalability barriers. The joint computation of LOF scores and graph convolutions imposes significant resource demands, limiting its practical deployment in latency-sensitive environments such as IoT security gateways or edge computing nodes.

To address these challenges, future work may incorporate scalable GCN variants such as GraphSAGE or FastGCN, which approximate neighborhood aggregation and reduce runtime complexity [45]. Graph sparsification or node sampling techniques may also be adopted to lower memory usage during inference. Additionally, the current model assumes a static graph during both training and deployment, which is not suitable for dynamically evolving traffic environments. Incorporating streaming graph construction and online learning will improve adaptability to real-time changes in network topology and attack surface.

In summary, while GCN-LOF provides a structurally principled and empirically validated anomaly detection framework, future extensions should prioritize interpretability, efficiency, and adaptability for deployment in dynamic and resource-constrained security infrastructures.

To complement this discussion with empirical evidence, we evaluated the system-level resource consumption of GCN-LOF during inference. The evaluation was conducted on a high-memory Linux server equipped with 512 GiB of RAM. As shown in Table 11, the model demonstrates low-to-moderate computational overhead in practice. CPU utilization averaged 21.1% with 78.6% idle time, while memory usage remained around 24 GiB, leaving more than 400 GiB of RAM available. Swap usage was moderate at 3.5 GiB, and approximately 49 GiB of buffer/cache memory was actively maintained. These observations indicate that GCN-LOF can be efficiently deployed in high-capacity servers or cloud environments, supporting large-scale network traffic analysis without requiring specialized hardware acceleration.

5.4. Propagation perspective on LOF-weighted aggregation

To better understand the role of LOF-based dynamic weights, we briefly analyze their influence on information propagation in GCN-LOF. Unlike standard GCNs that uniformly aggregate neighbor features, our model scales the message from node j by $(1 + \beta \cdot \text{LOF}(j))$, allowing structurally irregular nodes to exert stronger influence.

This design helps mitigate over-smoothing by preserving feature diversity around anomalous nodes, which is critical in sparse or imbalanced intrusion scenarios. The LOF score, as a proxy for local density deviation, introduces graph-topological awareness into the aggregation process [46]. As a result, the model becomes more responsive to isolated attack patterns that may be overlooked by uniform or similarity-only propagation schemes.

In addition, the weighted formulation remains bounded and stable, with hyperparameters α and β controlling the trade-off between feature similarity and anomaly emphasis. This ensures that anomaly signals are amplified without destabilizing the training process.

Overall, the LOF-weighted aggregation acts as a localized attention mechanism that integrates structural irregularity into graph learning, enhancing the model's ability to detect sparse and subtle anomalies.

5.5. Temporal adaptation and dynamic graph modeling

While GCN-LOF demonstrates strong detection capabilities across diverse traffic datasets, it currently relies on a static KNN-based graph structure constructed prior to model training. This static topology assumes fixed relationships among traffic samples and does not reflect the evolving nature of real-world network environments, where node interactions, traffic patterns, and attack behaviors change over time.

The use of a static graph limits the model's ability to adapt to newly emerging threats or shifting communication structures, potentially leading to outdated neighborhood contexts during inference. For example, reconnaissance scans or botnet behaviors may only manifest transiently and could be overlooked if the graph fails to update dynamically with new observations.

To improve adaptability and realism, future extensions of GCN-LOF could explore temporal or evolving graph construction strategies. One promising direction is the integration of streaming graph models, where edge structures are incrementally updated based on continuous traffic flow. Another is the adoption of dynamic graph neural networks, such as EvolveGCN or TGAT, which encode temporal dependencies directly into the learning process. These approaches would allow the model to track evolving topologies, preserve time-aware structural features, and enhance sensitivity to transient or stealthy anomalies.

Incorporating temporal dynamics into graph construction aligns more closely with the operational requirements of modern intrusion detection systems, particularly in environments characterized by rapid protocol shifts, attack bursts, or concept drift. We regard this as a valuable future improvement to further enhance the generalization and responsiveness of the proposed framework.

6. Conclusion

This paper presents GCN-LOF, a hybrid intrusion detection model that integrates GCN and the LOF to address the challenge of detecting sparse anomalies and modeling complex structural patterns in network traffic. By leveraging GCN's ability to extract global relational information and LOF's sensitivity to local density variations, the model enhances anomaly discrimination and improves robustness under dynamic and imbalanced network conditions.

Empirical evaluations conducted on four authoritative datasets—UNSW-NB15, CICIDS, Bot-IoT, and ToN-IoT—demonstrate that GCN-LOF consistently outperforms mainstream baseline methods, including IsoForest, DeepAE, EFC, and LOF. Notably, it achieves an average improvement of 5.3% in F1, while maintaining high accuracy and recall

across both structured and heterogeneous attack scenarios. These results validate the model's capacity to generalize to diverse network environments and confirm the scientific value of integrating local density estimation with graph-based feature propagation.

In future work, we aim to improve GCN-LOF's scalability and deployment efficiency through graph sampling, pruning, and parallel computation strategies tailored for edge computing environments. Additionally, the integration of cost-sensitive learning, online adaptation, and transfer learning will be explored to enhance model responsiveness to evolving threat landscapes. This research contributes a theoretically grounded and practically viable approach to advancing intelligent intrusion detection in cybersecurity applications.

CRedit authorship contribution statement

Zhenkai Qin: Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Lingying Li:** Writing – original draft, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Qining Luo:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Xiaochuan Yu:** Writing – review & editing, Visualization, Software. **Qian Zhang:** Writing – review & editing, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code used for the experiments in this study is openly accessible at: <https://github.com/bitbitlemon/GCN-LOF/tree/main>.

The datasets employed in this research can be obtained from the following repository: https://staff.itee.uq.edu.au/marius/NIDS_datasets/.

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